

Shower Device

Happy Campers

Element E: Application of STEM Principles and Practices

Dormitory residents and dormitory guests of our boarding school need a device to place their shower products. This is because they would want their shower products to be easily accessible in the communal shower. They would also like to do this without having a personal shower caddy and without setting it on the floor because it is unsanitary. The main stakeholders are the dormitory residents. They need a solution because the showers in the dormitory were not built with a solution to this problem and dormitory residents are not permitted to leave their shower products in the shower area. The only place to put a device in the dormitory showers are on a relatively narrow divider wall of less than an inch wide.

If we use ABS or PLA filament and keep the device off the ground, it will meet the needs and expectations of the stakeholders. ABS is a great waterproof filament for 3D printing according to all3dp.com. But due to our current resources, specifically the type of 3D printing filament, PLA will also be a successful type of filament to create our design out of to meet our waterproof requirements.

According to the University of Utah, fungal, viral, and bacterial infections can be caught while using communal showers if not taking the right precautions. For example, foot infections like athletes' feet can be contracted because of the moist, warm environment in the showers.

The average diameter of a shower bottle is about 3 inches. To meet the criteria of fitting 5 bottles on our shower device, we would need to have a surface area of 12.5 square inches. $5 \times 3 = 15$. Our prototype is designed to have dimensions of 12 inches wide and 9 inches deep. This

equates to a surface area of 108 square inches. $12 \times 9 = 108$. This meets our stakeholders needs because 5 or more bottles would fit on our prototype as well as an individual shower caddy.

The MASTAN2 application can help with structure analysis of the design to estimate if it can support enough weight. Also these can be the average weight of shower products to determine minimum load capacity.

We have also received feedback on how to improve our overall design. Critiques given to us are that we should remove the need to create a hole through the shower divider wall. Instead we should have the weight and balance of the device be solved by going over the shower divider wall instead of through it.

The experts we contacted to give us feedback responded and said our idea is good. Our use of STEM principles and design has been approved by multiple sources since we have executed the process of problems and priorities and created a shower device design based on them.

One of our contracted experts is qualified to review our solution because he is a STEM teacher and has a long history in the subject. Our other expert is a knowledgeable person in the aspect of 3D printing. He has worked with many different types of filament and completed multiple projects with a 3D printer.

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